

JISHNU S NAIR

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Summary

ML research engineer with 6+ years building production LLM and agent systems at ServiceNow. Currently working on RL post-training, world models for agents, and large-scale model training. Author on multiple papers at **tier-1 AI conferences**, with three issued/filed US patents.

Research Interests

RL post-training · World models for long-horizon agents · Large-scale LLM training and post-training infrastructure · Synthetic data and reward modeling · Evaluations and analysis · Agentic Orchestrations

Publications

- [1] **Do Enterprise Systems Need Learned World Models? The Importance of Context to Infer Dynamics**
[Jishnu Sethumadhavan Nair](#), Patrice Bechard, Rishabh Maheshwary, Surajit Dasgupta, et al.
NeurIPS 2026 arXiv:2605.12178
- [2] **EnterpriseOps-Gym: Environments and Evaluations for Stateful Agentic Planning and Tool Use in Enterprise Settings**
Shiva Krishna Reddy Malay, Shravan Nayak, [Jishnu Sethumadhavan Nair](#), Sagar Davasam, et al.
International Conference on Machine Learning (ICML), 2026 arXiv:2603.13594
- [3] **DNR Bench: Benchmarking Over-Reasoning in Reasoning LLMs**
Masoud Hashemi, Oluwanifemi Bamgbose, Sathwik Tejaswi Madhusudhan, [Jishnu Sethumadhavan Nair](#), et al.
AAAI Conference on Artificial Intelligence, 2026 (Oral) arXiv:2503.15793
- [4] **Apriel-Nemotron-15B-Thinker**
Shruthan Radhakrishna, Soham Parikh, Gopal Sarda, Anil Turkan, et al. (incl. [J. S. Nair](#))
arXiv preprint, 2025 arXiv:2508.10948

Experience

ServiceNow

March 2024 – Present

Senior Machine Learning Engineer

Hyderabad, India

- **World models for enterprise agents:** First-authored research on learned dynamics models for stateful enterprise agents, focusing on how context shapes an agent's ability to infer environment dynamics. (**NeurIPS 2026**)
- **RL post-training for agents:** Built RL post-training pipelines using **GRPO**, **DAPO**, and **PPO** on top of vLLM and DeepSpeed; tuned reward shaping and rollout collection for long-horizon agentic tasks.
- **Agentic environments and benchmarks:** Contributed to EnterpriseOps-Gym (**ICML 2026**) - environments and evaluations for stateful agentic planning and tool use - and to DNR Bench (**AAAI 2026**) for over-reasoning in reasoning LLMs.
- **Synthetic data infrastructure:** Designed a graph-based synthetic data pipeline in **LangGraph** powering SFT and RL training across multiple internal models.
- **Multi-agent orchestration:** Built a LangGraph-based orchestration framework used by sister teams for complex, multi-step enterprise workflows.
- **LLM evaluation:** Designed a holistic evaluation framework (LangChain) covering reasoning, instruction-following, and tool-use traits using both industry and internal datasets.

ServiceNow

September 2022 – February 2024

Machine Learning Engineer

Hyderabad, India

- **LLM fine-tuning:** Fine-tuned and evaluated **Qwen**, **Mistral**, **Llama**, **Pixtral** models for ServiceNow-specific summarization, multi-turn Q&A, and code generation.
- **Inference infrastructure:** Built an LLM inference pipeline on **vLLM**, **TGI**, and **TensorRT** with quantization and continuous batching, productionized across internal teams.
- **Similarity Search service:** Architected an enterprise-scale similarity search service using Transformer embeddings, ONNX, and **HNSW**; production system serving multiple downstream teams.

- **Clustering service:** Built a **FAISS**-backed KMeans clustering service with prefiltering, used for large-scale dataset organization across the platform.
- **Mentorship:** Mentored two interns through to successful project completion.

ServiceNow

March 2022 – August 2022

Software Engineer – AI/ML

Hyderabad, India

- **ML deflection:** Built an ML system identifying ChatBot-resolvable customer issues, deflecting **\$1.75M/quarter** in tickets from human resolution.

ServiceNow

July 2020 – February 2022

Associate Software Engineer – AI/ML

Hyderabad, India

- **Semantic search:** Built and deployed a RoBERTa-based semantic search service for incident-ticket KB retrieval, resolving **\$1.35M/year** of tickets without human intervention.
- **Virtual agent:** Built and maintained virtual agent topics using NLU.

DBS Partners

March 2019 – October 2019

Machine Learning Intern

Remote, India

- **Contract analytics:** NLP models for feature extraction from employment contracts (**RNN/LSTM, RASA NLU**).

Technical Skills

Languages: Python, SQL, Java, JavaScript

RL & Post-training: GRPO, DAPO, PPO, DPO, SFT, RLHF, reward modeling, TRL, Axolotl, Unsloth

LLMs & Agents: PyTorch, HuggingFace, DeepSpeed, vLLM, TensorRT, LangChain, LangGraph, ONNX

Training Infrastructure: Ray, H100/GPU clusters, NCCL, distributed training, Slurm, Hydra, Kubernetes

Evaluation & Tracking: LLM-as-Judge, benchmark design, synthetic data generation, Weights & Biases, MLflow

Serving & Storage: FastAPI, Docker, FAISS, Neo4j, Azure

Education

Government Model Engineering College, Kochi

2016 – 2020

Bachelor of Technology in Computer Science and Engineering

Kochi, India

CGPA : 8.8/10

Patents

Smart Persistence Of Model For Effective Predictions And Updates

January 2026

US Patent No. 18/626,139

Patent Issued

Platform Agnostic Scalable And High-performance Semantic Search Framework

September 2025

US Patent No. 18/623,844

Patent Issued

Framework Neutral And Updatable Clustering Model

April 2024

US Patent Application No. 18/426,810

Patent Filed

Awards

ATG Llama Award (2022, 2023): ServiceNow ML recognition; 14% award rate.